

THE TWO-PHOTON LASER

Daniel J. Gauthier and Hope M. Concannon

Department of Physics
Duke University
P.O. Box 90305
Durham, NC 27708
USA

ABSTRACT

We explain in simple terms the behavior of two-photon lasers and describe recent results that have lead to the realization of the first continuous-wave two-photon optical laser. We stress the differences between one- and two-photon lasers to develop an appreciation of their dynamics and the difficulties associated with achieving two-photon lasing. It will be shown that the fundamental light-matter interactions give them a unique threshold behavior and that competing nonlinear optical processes make it difficult to achieve lasing. Finally, we describe a new class of driven-atom gain media that are ideally suited for constructing two-photon lasers.

I. INTRODUCTION

Soon after the advent of the laser in the early 1960's, researchers discovered other types of quantum oscillators that have distinct and potentially useful characteristics. One such oscillator is known as the two-photon laser [1], so named because it is based on the two-photon stimulated emission process. The two-photon laser appeared to be an experimentalists dream because it was predicted [2] that it could produce two tunable frequencies, operate at high power and store large energies (recall that there were only fixed-frequency, modest power lasers in the early sixties). In addition, the two-photon laser challenged our understanding of the interaction of light with matter because it is a highly nonlinear, far from equilibrium system that cannot be analyzed using standard perturbation techniques. Unfortunately, the practical benefits of the laser have not been realized nor have the theories been tested because it has been difficult to achieve two-photon lasing [3] due to the lack of suitable gain media. In this chapter, we will explain in simple terms the behavior of two-photon lasers, discuss why it is difficult to

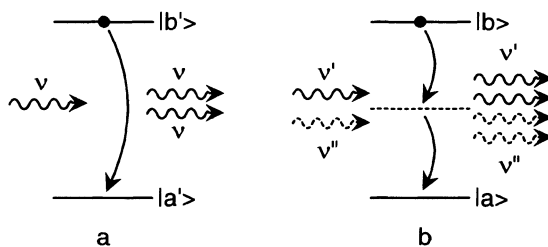


Figure 1 (a) The one-photon stimulated emission process. The states have opposite parity. (b) The two-photon stimulated emission process. The states have the same parity.

achieve two-photon lasing and describe recent work that led to the demonstration of the first continuous-wave two-photon optical laser.

An obvious question that might arise when reading the title of this chapter is: What is a two-photon laser? The answer to this question is quite simple if we compare and contrast the origin of gain in one- and two-photon lasers. In normal one-photon lasers, the gain is due to the one-photon stimulated emission process shown schematically in Fig. 1a. In this process, an incident photon can stimulate an atom from the upper state $|b'\rangle$ to the lower state $|a'\rangle$ and two photons are scattered by the atom. The scattered photons have the same frequency, phase and direction of the incident photon which give the laser its unique coherence properties. The stimulated emission rate is proportional to the incident photon flux and proceeds most efficiently when the frequency of the incident photon ω is equal to the transition frequency $\omega_{b'a'}$ and when the states have opposite parity (connected by an allowed electric dipole transition matrix element).

In contrast, the gain in two-photon lasers is due to the two-photon stimulated emission process shown schematically in Fig. 1b. In this process, two incident photons can stimulate an atom from the upper state $|b\rangle$ to the lower state $|a\rangle$ and four photons are scattered by the atom. Note that the frequencies ω' and ω'' of the incident photons can take on any value so long as $\omega' + \omega'' \simeq \omega_{ba}$, where ω_{ba} is the two-photon transition frequency and that the process proceeds most efficiently when the states have the same parity (not connected by an allowed electric dipole transition matrix element). The scattered photons have the same frequency, phase and direction of the incident photons which give the laser coherence properties different than those of normal one-photon lasers [4],[5]. Throughout the rest of the chapter, we assume for simplicity that there is only a single beam of light incident on the atom and hence that $\omega' = \omega'' \equiv \omega \simeq \omega_{ba}/2$. In this case, the two-photon stimulated emission rate is proportional to the square of the incident photon flux.

One consequence of the scaling of the two-photon stimulated emission rate with the incident photon flux is that the gain is intensity dependent. Consider a beam of light (intensity I_{IN}) incident on a collection of atoms that possesses a two-photon inversion. It will be amplified by an amount that is approximately given by

$$I_{OUT} \sim I_{IN} \exp(G^{(2)} I_{IN} L) \quad (1)$$

when the frequency of the incident beam is equal to one-half the two-photon transition

frequency, where L is the length of the medium and $G^{(2)}$ is the two-photon gain coefficient. In deriving Eq. 1, we have assumed that the transition is not saturated and that the gain is very small. Note that the input intensity appears in the exponential and hence the gain vanishes when the input intensity is small. This is in contrast to a one-photon gain medium where the gain is independent of the intensity (until saturation sets in).

The smallness of $G^{(2)}$ presents the main stumbling block to achieving two-photon lasing. Seemingly, a natural solution would involve using high intensities as can be seen from Eq. 1. Unfortunately, several other competing nonlinear optical processes occur simultaneously in the gain medium, such as four-wave mixing, harmonic generation, anti-Stokes Raman scattering, self-focusing, and photo ionization, and become more prevalent at higher intensities. In addition, normal one-photon lasing or superfluorescence from the upper state $|b\rangle$ to some other lower-energy intermediate state may deplete the inversion between the states $|b\rangle$ and $|a\rangle$. Typically, the competing processes dominate the two-photon stimulated emission process and prevent the occurrence of two-photon lasing.

Is there a two-photon gain medium that avoids these problems? To answer this question, it is necessary to develop a simple model that sets bounds on the operating parameters of the laser. This will quantify some of their differences from normal one-photon lasers and will focus our attention on the characteristics that must be optimized.

II. SIMPLE MODEL OF A TWO-PHOTON LASER

We have developed a model for two-photon lasers based on a set of self-consistent rate equations that predicts many of their crucial attributes without being overly complex. Although other models are required to make quantitative comparisons with experimental results, we feel that our model is useful for developing an intuitive understanding of two-photon lasers. Below, we present our model and use it to make predictions of the behavior of two-photon optical lasers. Later, we discuss effects ignored in our model and describe a new type of gain medium that was used to realize the first continuous-wave, two-photon optical laser.

A key test for any model of two-photon lasers is whether it predicts the novel threshold behavior of the laser. Briefly, the threshold condition for all lasers is that the round-trip gain must equal the round-trip loss. For one-photon lasers this criterion yields the well known result that lasing will commence when a uniquely defined minimum inversion (proportional to the gain) is attained via sufficient pumping. The situation is more complicated for the two-photon laser because the gain increases with increasing inversion density ΔN and with increasing cavity photon number q (until the atoms are saturated), so the threshold condition must be specified by two parameters. We define a threshold inversion density ΔN_{th} as the inversion needed to satisfy the threshold condition with a cavity photon number q_{sat} just sufficient to saturate the two-photon gain. When $\Delta N > \Delta N_{th}$, there is a corresponding cavity photon number (which is less than q_{sat}) that must be present in the cavity before the laser will turn on. Hence, if the laser is initially off it cannot turn on unless some perturbation, such as an externally injected field, brings it above the necessary value [1],[6],[7],[9],[10].

Our rate-equation model of the two-photon laser follows from the standard model of one-photon lasers with the exception that the one-photon stimulated emission rate

$$W^{(1)} = B^{(1)}q, \quad (2)$$

is replaced by the two-photon stimulated emission rate [11]

$$W^{(2)} = B^{(2)}q^2, \quad (3)$$

where $B^{(1)}$ ($B^{(2)}$) is the one- (two-) photon rate coefficient. The gain coefficient introduced in the previous section is given in terms of the rate coefficient through the relation

$$G^{(2)} = \frac{V_c V_a B^{(2)} \Delta N}{c^2 \hbar \omega}, \quad (4)$$

where V_c (V_a) is the volume of the cavity (volume of the medium contained in the cavity). For simplicity, we assume that the two-photon laser operates in the degenerate mode, that the laser oscillates in a single, plane-wave mode, and that the cavity (population) decay rate γ_c (γ) is much smaller than the atomic coherence dephasing rate. Under these oversimplifying conditions, the behavior of the laser is described by the mean photon number in the cavity q and the mean population inversion density ΔN between the atomic levels that participate in the stimulated emission process. The first-order, coupled nonlinear differential equations governing the evolution of these quantities are given by

$$\frac{dq}{dt} = V_a B^{(2)} q^2 \Delta N - \gamma_c (q - q_{inj}(t)), \quad (5)$$

and

$$\frac{d\Delta N}{dt} = -2B^{(2)} q^2 \Delta N - \gamma(\Delta N - \Delta N_o), \quad (6)$$

where ΔN_o is the inversion density in the absence of the field due to the pump process, $\gamma \Delta N_o$ is the pump rate, and $q_{inj}(t)$ is the photon number injected into the cavity by an external source. We see from Eq. 5 that the photon number increases due to the two-photon stimulated emission process and by injection from the external source, and decreases due to linear loss through the cavity mirrors. We have ignored the possibility of two-photon spontaneous emission processes at the laser frequency because the emission rates are extremely small in the optical regime [12]. This approximation is not valid for two-photon masers where the stimulated and spontaneous rates are comparable [13]. From Eq. 6, we see that the inversion decreases in response to the stimulated emission process and due to other radiative (at frequencies distinct from the laser frequency) and non radiative decay mechanisms, and increases due to the pump process.

The steady-state behavior of the two-photon laser can be obtained readily from our model. From Eq. 6, we find that

$$\Delta N_{ss} = \frac{\Delta N_o}{1 + q_{ss}^2/q_{sat}^2}, \quad (7)$$

where

$$q_{sat} = \sqrt{\gamma/2B^{(2)}} \quad (8)$$

is the two-photon saturation photon number. The saturation intensity is given in terms of the saturation photon number through the relation $I_{sat} = c \hbar \omega q_{sat}/V_c$. Equation 7 is reminiscent of the steady-state inversion for a one-photon laser [14] except that the denominator is not linear in q_{ss} . In contrast, the steady-state solution for the photon number is very different from that of one-photon lasers. We find three solutions given by

$$q_{ss}^o = 0, \quad (9)$$

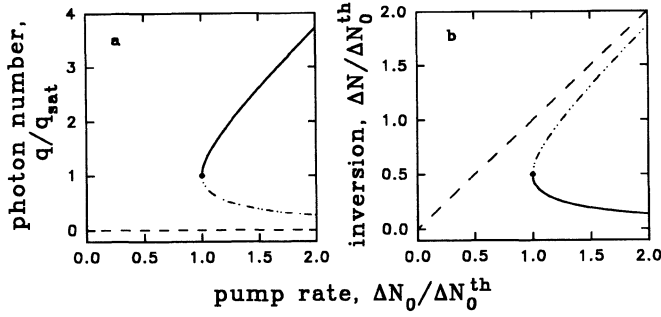


Figure 2 Steady state behavior of (a) the cavity photon number normalized to the saturation photon number and (b) the atomic inversion normalized to the threshold atomic inversion. The dashed, solid, and dot-dash lines are the $(q_{ss}^o, \Delta_{ss}^o)$, $(q_{ss}^+, \Delta_{ss}^+)$, and $(q_{ss}^-, \Delta_{ss}^-)$ solutions, respectively.

and

$$q_{ss}^{\pm} = \frac{\gamma V_a}{4\gamma_c} \left[\Delta N_o \pm \sqrt{\Delta N_o^2 - 16q_{sat}^2\gamma_c^2/\gamma^2V_a^2} \right], \quad (10)$$

for the case when $q_{inj}(t) = 0$. The physically meaningful (real) steady-state solutions represented by these equations are shown in Fig. 2 where they are plotted as a function of the pump rate. It is seen from Eq. 10 that

$$\Delta N_o^{th} = \frac{4q_{sat}\gamma_c}{\gamma V_a} \quad (11)$$

is the minimum value of ΔN_o which admits a nonzero photon number. This yields $\Delta N_{ss} = 1/2 \Delta N_o^{th}$ and $q_{ss}^{\pm} = q_{sat}$ at threshold, in agreement with the heuristic discussion of the threshold behavior presented earlier.

The discontinuous threshold behavior shown in Fig. 2 is indicative of a first-order phase transition, which is very different from the smooth turn-on behavior of normal one-photon lasers. Note from Fig. 2a that the two-photon gain is saturated at threshold, again in sharp contrast to the typical one-photon laser which operates very far below saturation. Figure 2b also shows that the inversion is never constant, unlike the behavior of one-photon lasers where the inversion clamps above threshold [14]. Rate equation models that assume the inversion remains constant have been investigated previously [10].

Based on past experience with one-photon lasers, we expect that the steady-state solutions may be unstable because the laser operates in the saturated regime. We have performed a linear stability analysis of the steady-state solutions and find that: 1) the zero-photon solution $(q_{ss}^o, \Delta_{ss}^o)$ is always stable (two-photon spontaneous emission, neglected here, can destabilize this solution); 2) the $(q_{ss}^-, \Delta_{ss}^-)$ solution, where the photon number decreases with increasing pump rate, is always unstable; 3) the $(q_{ss}^+, \Delta_{ss}^+)$ solution is always stable for a 'good' cavity ($\gamma/\gamma_c > 1$); and 4) the $(q_{ss}^+, \Delta_{ss}^+)$ solution is unstable for a 'bad' cavity ($\gamma/\gamma_c < 1$) for pumping just above threshold but stabilizes for higher pump rates [6], [7]. Ning and Haken [15] have found that the laser is always unstable for a bad cavity when coherent effects are taken into account.

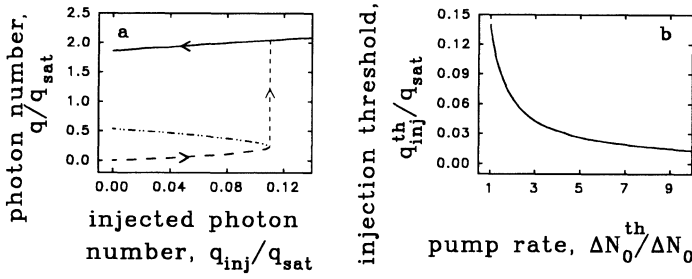


Figure 3 The injection threshold. (a) Steady-state photon number with a cw field injected into the cavity for a pump rate 20% above threshold. Threshold is defined as the point where the lowest solution ends. The arrows indicate the progression of the laser from off to on. (b) Variation of the injection threshold with the pump rate.

To address how an injected field can turn on the laser, we have determined the steady-state solutions of Eqs. 5 and 6 and their stability properties when a continuous-wave beam of photons is injected into the cavity ($q_{\text{inj}} \neq 0$). The stability of the solutions can be quite complex, especially in the bad-cavity limit. However, the analysis is straightforward if we consider the important case when the laser is initially ‘off’, $\Delta N_o > \Delta N_o^{\text{th}}$, and q_{inj} is increased slowly. This behavior is illustrated in Fig. 3a where we plot the three solutions for the steady-state photon number when $q_{\text{inj}} \neq 0$ and $\Delta N_o = 1.2 \Delta N_o^{\text{th}}$. The low-power solution (dashed curve) increases as q_{inj} increases and is stable until it reaches a critical value $q_{\text{inj}}^{\text{th}} \simeq 0.11 q_{\text{sat}}$. This point defines the minimum injected photon number necessary to initiate lasing. Increasing q_{inj} beyond $q_{\text{inj}}^{\text{th}}$ forces the laser to switch from the low-power solution (now unstable) to the high-power solution (stable); the laser turns on. The behavior of the laser as q_{inj} decreases depends on the quality of the cavity. For a good cavity, the laser will continue to operate at high power as q_{inj} decreases. For a bad cavity, the laser will only continue to operate at high power if the solution is stable when $q_{\text{inj}} = 0$. We have studied the stability behavior for other values of the pump rate and find that the injection threshold $q_{\text{inj}}^{\text{th}}$ decreases for higher values of the pump rate as shown in Fig. 3b. In practice, a pulse (peak photon number q_{inj}^o) is injected into the laser rather than a cw beam. When the laser can (cannot) adiabatically follow the temporal variation of the pulse, $q_{\text{inj}}^o = q_{\text{inj}}^{\text{th}}$ ($q_{\text{inj}}^o > q_{\text{inj}}^{\text{th}}$).

The transient behavior of the laser described above is explored by numerically integrating Eqs. 5 and 6. Figure 4 shows how the laser responds to injected trigger pulses for a good cavity ($\gamma/\gamma_c = 2$) when the pump rate is greater than the threshold pump rate ($\Delta N_o = 1.2 \Delta N_o^{\text{th}}$) and when there are no photons in the cavity initially. For a weak trigger pulse (peak amplitude $q_{\text{inj}}^o = 0.1 q_{\text{sat}}$, Fig. 4a) the laser is not driven above threshold, while for a slightly stronger pulse (peak amplitude $q_{\text{inj}}^o = 0.12 q_{\text{sat}}$, Fig. 4b) the laser is driven above threshold and it attains a constant amplitude after the injected pulse switches off. This is in good agreement with the injection threshold $q_{\text{inj}}^{\text{th}} \simeq 0.11 q_{\text{sat}}$ calculated above.

The most important conclusion that we can draw from our model is that the intensity circulating in the laser resonator is always greater than or equal to the two-photon

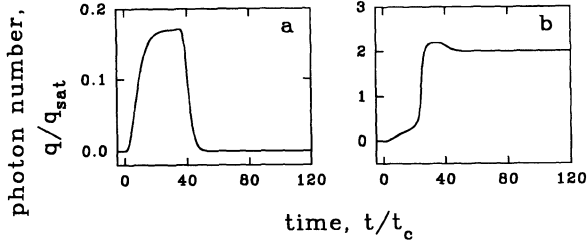


Figure 4 Transient evolution of the photon number for pumping 20% above threshold. (a) $q_{inj} = 0.1$ The trigger is not strong enough to reach threshold. (b) $q_{inj} = 0.12$ The laser is driven to the high-power state.

saturation intensity. Hence, the gain medium must have a low two-photon saturation intensity to avoid the high intensities that tend to magnify competing nonlinear optical processes. Fortunately, low saturation intensities go hand-in-hand with large two-photon gains as can be seen from Eqs. 4 and 8.

III. TWO-PHOTON GAIN IN A THREE-LEVEL ATOMIC SYSTEM

To estimate the size of the two-photon gain and saturation intensity, we will investigate the microscopic interaction of a field with a three-level atomic system using the density-matrix formalism developed in earlier chapters. Consider a field $\vec{E}(t) = [\vec{E}_0 \exp(-i\omega t) + c.c.]$ interacting with the inverted $|b\rangle \rightarrow |a\rangle$ two-photon transition as shown in Fig. 5. Since these states have the same parity, they are not connected by a normal one-photon transition and hence the electric-dipole matrix element is zero. A real intermediate level $|i\rangle$ with opposite parity is located near the virtual intermediate level of the two-photon transition (detuning Δ_{ia}) to enhance the two-photon transition rate. The electric-dipole matrix element of the $|b\rangle \rightarrow |i\rangle$ ($|i\rangle \rightarrow |a\rangle$)

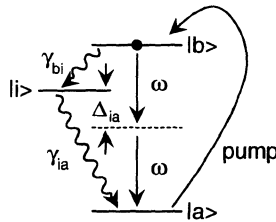


Figure 5 Two-photon gain in a three-level atomic system. The intermediate state $|i\rangle$ enhances the two-photon transition rate.

transition is denoted by $\vec{\mu}_{bi}$ ($\vec{\mu}_{ia}$) and the corresponding Rabi frequency is denoted by $\Omega_{bi} = 2\vec{\mu}_{bi} \cdot \vec{E}_o/\hbar$ ($\Omega_{ia} = 2\vec{\mu}_{ia} \cdot \vec{E}_o/\hbar$). Population in the upper level can decay spontaneously to the intermediate level (rate γ_{bi}) which subsequently decays to the lower level (rate γ_{ia}). For simplicity, we assume that $\gamma_{ia} \gg \gamma_{bi}$ so that essentially no population builds up in the intermediate level.

The only nonzero elements of the density matrix are: the populations ρ_{aa} and ρ_{bb} ; the one-photon coherences ρ_{bi} and ρ_{ia} ; and the two-photon coherence ρ_{ba} . The equations-of-motion for the density matrix are obtained from Heisenberg's equation and are simplified using the one- and two-photon rotating-wave approximations. In addition, the equations are simplified by adiabatic elimination of the one-photon coherences (performed formally by setting $\dot{\rho}_{bi} = \dot{\rho}_{ia} = 0$) which is valid so long as $\Delta_{ia} \gg \gamma_{bi}$, Ω_{bi} , Ω_{ia} [17]. After some algebra, it is found that the equations-of-motion for the population inversion density $\Delta N = \eta(\rho_{bb} - \rho_{aa})$ and the slowly-varying two-photon coherence density $\sigma_{ba} = \eta \rho_{ba} \exp(i2\omega t)$ are given by

$$\frac{d\Delta N}{dt} = -i[(\Omega^{(2)})^* \sigma_{ba} - \sigma_{ab} \Omega^{(2)}] - \gamma(\Delta N - \Delta N_o), \quad (12)$$

and

$$\frac{d\sigma_{ba}}{dt} = -i \frac{\Delta N \Omega^{(2)}}{2} - [i(\Delta_2 - \Delta_s) + \Gamma_{ba}] \sigma_{ba}, \quad (13)$$

where η is the atomic number density, $\gamma \simeq \gamma_{bi}$ is the inversion decay rate, $\Delta_2 = 2\omega - \omega_{ba}$ is the two-photon detuning, Γ_{ba} is the two-photon coherence dephasing rate, $\Omega^{(2)} = \Omega_{bi}\Omega_{ia}/\Delta_{ia}$ is the two-photon Rabi frequency, and $\Delta_s = (|\Omega_{ia}|^2 - |\Omega_{ia}|^2)/\Delta_{ia}$ is the AC Stark shift of the two-photon transition. Note that Eqs. 12 and 13 are nearly identical to the corresponding equations for a driven two-level atom. It can be seen from Eq. 13 that the two-photon transition frequency is intensity-dependent due to the AC Stark shift. These shifts can play an important role in the dynamics of two-photon lasers.

The results obtained from the density matrix formalism are connected to the rate equations used in the previous section under conditions when there is large dephasing of the two-photon coherence ($\Gamma_{ba} \gg \Delta_2, \Delta_s, \gamma, \gamma_c$). In this case, the coherence can be adiabatically eliminated from Eq. 12 using Eq. 13 with $\dot{\sigma}_{ba} = 0$. By comparison with Eq. 6, it is found that the two-photon rate coefficient is given by

$$B^{(2)} = \frac{32\pi^2 |\vec{\mu}_{bi} \cdot \hat{\epsilon}|^2 |\vec{\mu}_{ia} \cdot \hat{\epsilon}|^2 \omega^2}{V_c^2 \hbar^2 \Delta_{ia}^2 \Gamma_{ba}}, \quad (14)$$

where $\hat{\epsilon}$ is the polarization unit-vector of the field. From Eq. 14, it is straightforward to determine the saturation photon number using Eq. 8 and the threshold inversion using Eq. 11. Note that the rate coefficient can be enhanced significantly when the real intermediate level $|i\rangle$ is close to the virtual level of the two-photon process, that is, when Δ_{ia} is small. It was an understanding of this scaling that led to the development of the first microwave two-photon oscillator [13],[18] and the first two-photon optical laser [23].

What are numerical values for these parameters? Evaluating Eqs. 14, 8, and 11 using values that are typical for a two-photon optical gain medium (see Table 1) and a resonator with a loss-per-pass of 1%, we find that $B^{(2)} = 2.6 \times 10^{-12} \text{ s}^{-1}$, $q_{sat} = 4.4 \times 10^9$ photons (corresponding to $I_{sat} = 6.6 \times 10^4 \text{ W-cm}^{-2}$), and $\Delta N_o^{th} = 1.8 \times 10^{13} \text{ atoms-cm}^{-3}$. We see that the intensity circulating in the resonator is modest in comparison to some present-day lasers, giving hope that competing nonlinear optical effects can be avoided (such as self-focusing and photo ionization). In addition, we see that the

Table 1 Parameters used to calculate the two-photon rate coefficient.

dipole moment, $ \vec{\mu}_{bi} \cdot \hat{\epsilon} = \vec{\mu}_{ia} \cdot \hat{\epsilon} $	$ea_o = 2.4 \times 10^{-18} \text{cm}^{5/2} \text{g}^{1/2} \text{s}^{-1}$
population decay rate, γ	10^8s^{-1}
coherence dephasing rate, Γ_{ba}	$1 \times 10^9 \text{s}^{-1}$
intermediate level detuning, Δ_{ia}	$1 \times 10^{13} \text{s}^{-1}$
laser frequency, ω	$3 \times 10^{15} \text{s}^{-1}$
cavity length, L	5 cm
cavity volume, V_c	$6 \times 10^{-4} \text{cm}^3$
cavity decay rate, γ_c (corresponds to 1% loss)	$6 \times 10^7 \text{s}^{-1}$

threshold inversion density is quite low, so it should be straightforward to achieve two-photon lasing.

Unfortunately, nature is not so kind! We must consider the possibility that normal one-photon lasing can occur at a spectrally distinct frequency on the $|b\rangle \rightarrow |i\rangle$ transition. For sufficiently low one-photon gains, one-photon lasing can be avoided using a high-finesse cavity that selectively enhances the two-photon laser frequency and not the one-photon laser frequency. For high one-photon gain, superfluorescence will occur which will effectively destroy the inversion. The threshold for superfluorescence occurs when the single-pass one-photon gain $G^{(1)}L \sim 30$ which can be avoided only by reducing the number density (which also reduces the two-photon gain!). We estimate the single-pass one-photon gain on the $|b\rangle \rightarrow |i\rangle$ transition using

$$G^{(1)}L = \frac{4\pi\Delta N_o^{th} |\vec{\mu}_{bi} \cdot \hat{\epsilon}|^2 \omega L}{\hbar c \Gamma_{ba}} \quad (15)$$

and find that $G^{(1)}L = 640$. Hence the number density must be reduced by a factor of ~ 20 to just avoid superfluorescence and by a factor of ~ 100 to avoid significant depletion of the inversion. For a $100\times$ decrease in the number density, the loss-per-pass in the resonator must be less than $\sim 10^{-4}$ to achieve two-photon lasing. Such low-loss resonators can be constructed using recently developed super-polished mirrors. Note, however, that the gain medium must be in contact with the mirrors that form the resonator because any other optics in the cavity will give too high of a loss.

IV. DRESSED-STATE TWO-PHOTON GAIN MEDIUM

One obvious way to increase the two-photon gain on the $|b\rangle \rightarrow |a\rangle$ transition without increasing the one-photon gain on the $|b\rangle \rightarrow |i\rangle$ transition is to decrease the detuning Δ_{ia} from the intermediate transition $|i\rangle$. Two-photon lasing can be achieved using a resonator with 1% loss by decreasing Δ_{ia} by a factor of 100. In this case, $B^{(2)} = 2.6 \times 10^{-8} \text{s}^{-1}$, $q_{sat} = 4.4 \times 10^7$ photons (corresponding to $I_{sat} = 6.6 \times 10^2 \text{W-cm}^{-2}$), and $\Delta N_o^{th} = 1.8 \times 10^{11} \text{atoms-cm}^{-3}$.

The detuning from the intermediate level is not a parameter that can be adjusted easily, however. The atomic energy level structure is fixed and we know of no system in the optical regime that has such a small detuning. The breakthrough in the development of two-photon optical lasers occurred when Prof. Mossberg's group at the University of Oregon realized that it is possible to 'engineer' a near-ideal two-photon

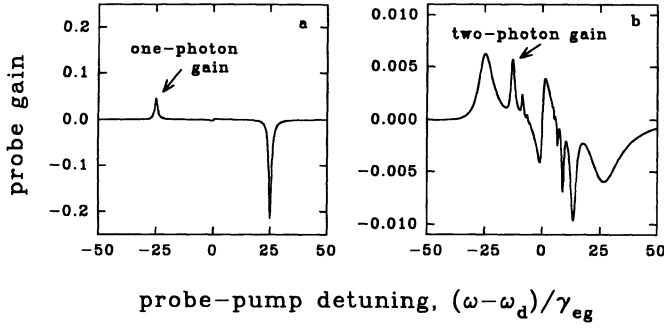


Figure 6 The response of a driven two-level atom with $\Omega'_d/\gamma_{eg} = 25$ and $\Delta_d = -0.4\Omega_d$ (a) The gain experienced by weak probe beam. (b) The gain experienced by a strong probe beam.

gain medium that possesses a small and adjustable detuning [19]. It consists of a two-level atom (one-photon transition) driven by an intense, near-resonant laser field. In the rest of this section, we will briefly describe the origin of two-photon gain in the driven-atom system.

Consider a two-level atom with a ground (excited) level denoted by $|g\rangle$ ($|e\rangle$) driven by an intense laser field $\vec{E}_d(t) = [\vec{E}_d \exp(-i\omega_d t) + \text{c.c.}]$. We are to think of the laser-driven atom as a composite system (a ‘dressed’ atom) rather than separate entities. The atomic states have opposite parity and are connected by an allowed electric-dipole moment $\vec{\mu}_{eg}$. The population decay rate (coherence dephasing rate) is denoted by γ_{eg} (Γ_{eg}) and we assume for simplicity that the system experiences only natural broadening so that $\Gamma_{eg} = \gamma_{eg}/2$. The interaction strength of the driven-atom system is characterized by the generalized Rabi frequency $\Omega'_d = \sqrt{\Omega_d^2 + \Delta_d^2}$, where $\Omega_d = 2\vec{\mu}_{eg} \cdot \vec{E}_d/\hbar$ is the resonant Rabi frequency and $\Delta_d = \omega_d - \omega_{eg}$ is the detuning of the driving field from the atomic resonance frequency ω_{eg} . The Rabi frequency can be of the order of $6 \times 10^9 \text{ s}^{-1}$ for a strong optical transition and a 1 W laser beam collimated to a diameter of $\sim 1 \text{ mm}$.

It has been known for quite some time that the absorption spectrum of a weak probe beam interacting with a collection of dressed atoms has a spectrally narrow gain feature for nonzero atom-driving field detunings [20]. The gain feature shown in Fig. 6a occurs at a probe beam frequency $\omega = \omega_d - \Omega'_d$ for $\Delta_d < 0$, its width is equal to the line width of the undriven transition Γ_{eg} and the maximum gain is equal in magnitude to $\sim 0.05\alpha_{eg}$, where α_{eg} is the line-center absorption coefficient of the undriven system. Most importantly, the strength of the gain is independent of the phase of the probe field; hence, the gain is not due to some type of parametric wave-mixing process. Rather, it is essentially identical to the familiar one-photon gain medium (several researchers have successfully constructed one-photon lasers using dressed atoms [21]).

The absorption spectrum of a dressed atom becomes complex when the probe beam intensity increases as shown in Fig. 6b. Several new, intensity-dependent gain features appear that occur at subharmonics of the generalized Rabi frequency, that is, for probe beam frequencies $\omega = \omega_d - \Omega'_d/n$ ($n = 1, 2, 3, \dots$). The feature that occurs at $\omega =$

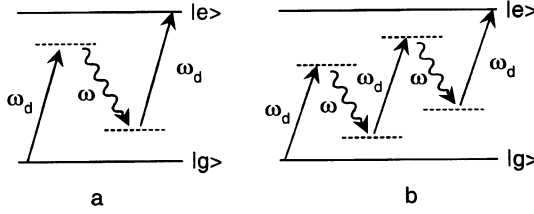


Figure 7 Perturbation-theory interpretation of the gain in a strongly driven atom. Hyper-Raman processes give rise to (a) one-photon and (b) two-photon gain in the driven-atom system.

$\omega_d - \Omega_d/2$ arises from two-photon stimulated emission and was used to realize the first two-photon optical laser. The fact that this is a two-photon gain feature was not fully appreciated until recently - the early studies called it a ‘subharmonic resonance’.

A crude understanding of the origin of the gain features can be obtained from a perturbation-theory analysis of the interaction of the probe beam and the driven atom. Of course, the analysis cannot fully explain the interaction because the pump beam is far too intense to be treated perturbatively. It is found that the one- and two-photon gain features arise from a type of hyper-Raman scattering process as shown in Fig. 7. In both processes, probe beam photons are created at the expense of pump beam photons when the fields induce a transition from the ground level $|g\rangle$ to the excited level $|e\rangle$. Hence, the ‘upper’ (‘lower’) laser level is actually the ground (excited) level of the atom for the driven-atom gain medium. The ‘inversion’ between the upper and lower laser levels is maintained by spontaneous emission from $|e\rangle \rightarrow |g\rangle$. Note that there is never an inversion between the excited and ground levels of the atom.

A quantitative (and intuitive) understanding of the origin of the gain features can be obtained using the dressed-state interpretation of the driving-field - atom interaction. The analysis properly accounts for the interaction of the intense pump field with the atom and treats as perturbations the probe field and spontaneous emission. We note that the study of driven two-level atoms transcends our present application of two-photon gain; it is a textbook example of the application of quantum electrodynamics and so we will only sketch the development of the theory. Our notation follows closely the recent and well-written text by Profs. Cohen-Tannoudji, Dupont-Roc, and Grynberg [22].

The dressed-state analysis differs from the standard semi-classical analysis in that the field mode of the driving field is treated quantum mechanically rather than classically. The reduced Hamiltonian for the driving-field - atom system can be broken up into ‘bare’ atom, ‘bare’ field, and interaction parts and is denoted by

$$\hat{H}_R = \hat{H}_{atom} + \hat{H}_{driving\ field} + \hat{V}_{interaction} . \quad (16)$$

The two-level atom Hamiltonian is given by

$$\hat{H}_{atom} = \hbar\omega_{eg} |e\rangle\langle e| \quad (17)$$

which has eigenstates $\{|g\rangle, |e\rangle\}$ and eigenenergies $\{0, \hbar\omega_{eg}\}$. The driving-field

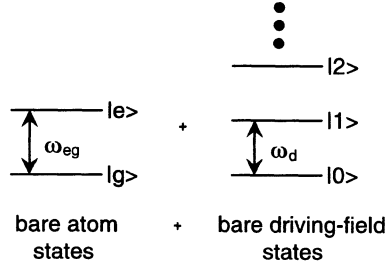


Figure 8 Energy eigenstates of the (a) bare atom and (b) the bare driving field.

Hamiltonian is given by

$$\hat{H}_{driving\ field} = \hbar\omega_d(\hat{a}_d^\dagger\hat{a}_d + \frac{1}{2}) \quad (18)$$

which has eigenstates $\{|n\rangle\}$ and eigenenergies $\{\hbar\omega_d(n + 1/2)\}$, where $\hat{a}_d$ ($\hat{a}_d^\dagger$) is the photon annihilation (creation) operator and n is the photon occupation number (the number of photons in the mode). The energy eigenstates of the uncoupled atom and field are shown schematically in Fig. 8a and 8b. It is seen that the driving-field eigenstates form an infinite ladder of equally spaced levels which is the underlying reason that the driven-atom system is a good two-photon gain medium.

The eigenstates of $\hat{H}_{atom} + \hat{H}_{driving\ field}$ are just the product states formed from the ‘bare’ atom and field states $\{|g, n\rangle, |e, n\rangle\}$, which are shown schematically in Fig. 9a. It is seen that the states $|g, n+1\rangle$ and $|e, n\rangle$ are close in energy for a near-resonant driving field and are degenerate for resonant driving. The interaction part of the Hamiltonian couples the nearly degenerate levels and removes any degeneracies in the system. It is given by

$$\hat{V}_{interaction} = -\hat{\vec{\mu}} \cdot \hat{\vec{E}}_d = -\hbar g_d(|e\rangle\langle g| + |g\rangle\langle e|)(\hat{a}_d + \hat{a}_d^\dagger), \quad (19)$$

where

$$\hat{\vec{\mu}} = \vec{\mu}_{eg}(|e\rangle\langle g| + |g\rangle\langle e|) \quad (20)$$

is the dipole-moment operator,

$$\hat{\vec{E}}_d = \tilde{\epsilon}_d \sqrt{\frac{8\pi\hbar\omega_d}{V_d}}(\hat{a}_d + \hat{a}_d^\dagger) \quad (21)$$

is the electric field operator, and

$$g_d = \vec{\mu}_{eg} \cdot \tilde{\epsilon}_d \sqrt{\frac{8\pi\omega_d}{\hbar V_d}} \quad (22)$$

is the atom-field coupling constant. The polarization unit vector of the driving field is denoted by $\tilde{\epsilon}_d$ and the volume of the driving-field mode is denoted by V_d . We have assumed, for simplicity, that $\vec{\mu}_{eg}$ is real. The nearly-degenerate product states are the only levels coupled by $\hat{V}_{interaction}$ where the interaction energies are given by

$$\langle e, n | \hat{V}_{interaction} | g, n+1 \rangle = \hbar g_d \sqrt{n+1}. \quad (23)$$

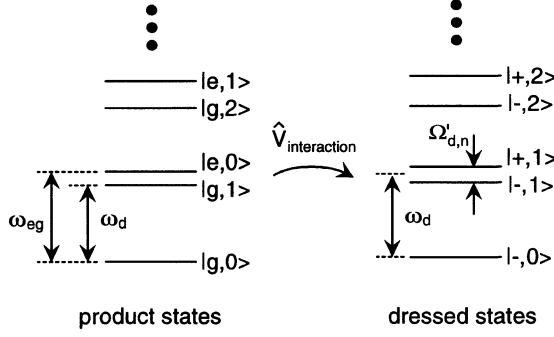


Figure 9 Energy eigenstates of (a) the uncoupled atom-field system (the product states), and (b) the coupled system (the dressed states).

While it is possible to solve for the dynamics of the interaction using the product-state basis, it is somewhat inconvenient because there are off-diagonal matrix elements of H_R .

A more natural approach is to use the dressed-state basis, that is, the basis formed by the energy eigenstates $\{|-,n\rangle, |+,n\rangle\}$ of H_R . They are a linear combination of the product states and are given by

$$\begin{aligned} |-,n\rangle &= \cos \theta_n |g,n\rangle - \sin \theta_n |e,n-1\rangle \\ |+,n\rangle &= \sin \theta_n |g,n\rangle + \cos \theta_n |e,n-1\rangle \end{aligned} \quad , \quad (24)$$

where the rotation angle is defined through $\tan 2\theta_n = -g_d\sqrt{n}/\Delta_d$. The dressed states are shown schematically in Fig. 9b. We see that there is an infinite ladder of levels where each rung of the ladder is a doublet of levels. The doublet splitting is a function of the photon occupation number and is given by

$$\Omega'_{d,n} = \sqrt{n g_d^2 + \Delta_d^2} . \quad (25)$$

We now explore ways in which the infinite-dimensional problem can be simplified before considering the interaction of the dressed atom with the other modes of the electromagnetic field (including the probe field). The dimensionality of the problem can be reduced dramatically by including in the calculation the dressed levels that are selectively occupied by the driving field rather than the complete set of states. For a driving field produced by an intense laser (properly described quantum mechanically by a coherent state), only a narrow distribution of dressed levels high up the ladder are occupied. We say that the levels are ‘locally periodic’ since the relative populations and the splitting of a dressed-state doublet are essentially the same independent of the position on the ladder (that is, the value of n). In this case $g_d\sqrt{n} \sim g_d\sqrt{\bar{n}} \equiv \Omega_d$, $\tan 2\theta_n = -\Omega_d/\Delta_d$, and $\Omega'_{d,n} = \sqrt{\Omega_d^2 + \Delta_d^2} \equiv \Omega'_d$, where $\bar{n}$ is the mean photon number and Ω_d is the classical Rabi frequency of the field (a constant).

The interaction between the dressed atom and the other modes of the field can be treated straightforwardly when there is local periodicity because we only need to

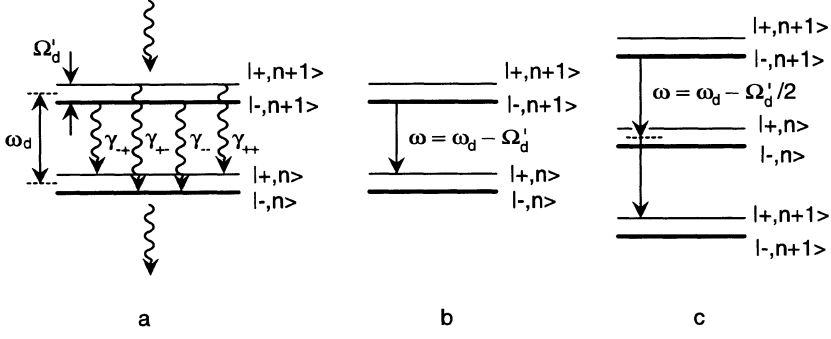


Figure 10 (a) The relative populations of the dressed-atom doublets become unbalanced due to spontaneous emission. One- (b) and (c) two-photon gain in the dressed-atom system.

consider transitions induced between one rung on the ladder and its nearby neighbors by the perturbation. The full dressed-atom - field interaction is just the sum of the local interactions. As usual, the strength of the interaction is related to the electric-dipole matrix elements between the dressed levels. It is found that all matrix elements are zero except for those that connect one rung of the ladder with its nearest neighbor. They are defined through the relation

$$\vec{\mu}_{ij}^{\pm} = \langle i, n \pm 1 | \hat{\vec{\mu}} | j, n \rangle \quad i, j = +, -. \quad (26)$$

where $\vec{\mu}_{ij}^+ = (\vec{\mu}_{ij}^-)^*$. For example, it is found that $\vec{\mu}_{+-}^+ = \vec{\mu}_{eg} \cos^2 \theta$. Note that the matrix elements are a function of the driving-field Rabi frequency and detuning through the rotation angle θ .

Coupling the dressed atom to all the unoccupied field modes leads to spontaneous decay of the levels. Considering the selection rules (Eq. 26), we see that each level on a rung of the ladder decays to both levels one step down the ladder at a rate

$$\gamma_{ij} = \gamma_{eg} \frac{|\vec{\mu}_{ij}^+|^2}{|\vec{\mu}_{eg}|^2} \quad i, j = +, -. \quad (27)$$

For example, it is found that $\gamma_{+-} = \gamma_{eg} \cos^4 \theta$. After several spontaneous emission events, the relative populations of a dressed-state doublet reach equilibrium in response to spontaneous decays into and out of the doublet. The relative population distribution is independent of n due to the local periodicity and is given by

$$\frac{\rho_{++}}{\rho_{--}} = \tan^4 \theta. \quad (28)$$

When the driving field is on resonance ($\Delta_d = 0$), both levels are equally populated since $\theta = \pi/4$. For off-resonance driving the populations become unbalanced. For example, $\rho_{--} > \rho_{++}$ when $\Delta_d < 0$ as shown in Fig. 10a.

A probe beam (frequency ω) interacting with the dressed atom will experience gain or absorption when the populations of the dressed levels are unbalanced just as a bare

atom will absorb or amplify a beam of light. It can be seen from Fig. 10b that a probe beam interacting with the $|+, n \rangle \rightarrow |-, n+1\rangle$ transition will be amplified due to one-photon stimulated emission when $\omega = \omega_d - \Omega'_d$ for $\Delta_d < 0$. This is precisely the spectral location of the gain feature shown in Fig. 6a. Analogously, one-photon absorption will occur on the $|-, n \rangle \rightarrow |+, n+1\rangle$ transition.

An intense probe beam can induce multiphoton transitions among the dressed levels. In particular, a probe beam interacting with the $|-, n+1\rangle \rightarrow |+, n-1\rangle$ will be amplified due to two-photon stimulated emission when $\omega = \omega_d - \Omega'_d/2$ as shown in Fig. 10c. Most importantly, the $|+, n\rangle$ and $|-, n\rangle$ intermediate levels are nearly degenerate with the virtual level of the two-photon transition thereby enhancing significantly the two-photon rate coefficient. In addition, the two-photon gain is spectrally distinct from the regions of large one-photon gain; hence, a high-finesse resonator can selectively enhance the two-photon gain. It is precisely for these reasons that dressed atoms make a near ideal two-photon gain medium. Note that the driven-atom system can support n -photon gain due to n -photon stimulated emission from one rung on the ladder to n rungs below.

The two-photon rate coefficient can be determined using the density-matrix formalism and by treating the probe as a classical field (similar to the analysis presented above in Sec. III). Obviously, the local dynamics of the dressed levels is richer than the dynamics of the simple three-level system considered in Sec. III because there are two intermediate levels. One complication that arises from the two intermediate states is that there are two quantum mechanical pathways from the excited level $|-, n+1\rangle$ to the lower level $|+, n-1\rangle$. The two pathways are taken into account by adding the contributions from each pathway to the induced polarization. A somewhat more subtle effect is the presence of coherences between the intermediate levels. Under conditions when $\Omega'_d \gg \Gamma_{eg}$, these coherences have a negligible contribution and can be ignored (this is known as the secular approximation). From the detailed calculations, it is found that the rate coefficient is a complex function of the driving-field detuning and is maximized for a given Ω_d when $\Delta_d \sim -0.4\Omega_d$. At the optimum detuning

$$B^{(2)} \sim \frac{3\pi^2 |\vec{\mu}_{eg} \cdot \hat{\epsilon}|^4 \omega^2}{8V_c^2 \hbar^2 (\Omega'_d/2)^2 \Gamma_{eg}}. \quad (29)$$

We see that the dressed-state and three-level atom rate coefficients are similar with $\Delta_{ia} \rightarrow \Omega'_d/2$ by comparing Eqs. 29 and 14. Thus, the rate coefficient for a dressed atom can be considerably larger than a typical three-level atom because Ω'_d can be small. Note that there is an optimum value of Ω'_d for a particular experimental situation since the one- and two-photon gain features will start to overlap as Ω'_d becomes small. Using $\Omega'_d = 6 \times 10^9 \text{ s}^{-1}$ and the parameters listed in Table 1, we find that $B^{(2)} \sim 3.4 \times 10^{-7} \text{ s}^{-1}$, $q_{sat} = 1.2 \times 10^7 \text{ photons}$ (corresponding to $I_{sat} = 1.8 \times 10^2 \text{ W-cm}^{-2}$), and $\Delta N_o^{th} = 4.8 \times 10^{10} \text{ atoms-cm}^{-3}$. Most importantly, we estimate that the one-photon dressed-atom gain that occurs at $\omega = \omega_d - \Omega'_d$ is only $G^{(1)}L \sim 0.1$. Therefore, it should be possible to realize a dressed-atom two-photon laser using a high-finesse resonator that selectively enhances the two-photon transition.

V. THE DRESSED-STATE TWO-PHOTON LASER

The experiments that were conducted to test the theoretical discussion outlined in the previous section used a unique experimental apparatus to mimic as closely as possible the theoretical assumptions [23],[24]. The 553.5-nm $^1S_o \rightarrow ^1P_1$ transition of

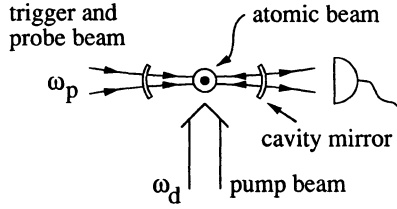


Figure 11 Schematic of the two-photon laser.

barium (natural linewidth $\gamma_{eg}/2\pi = 19$ MHz) approximated the two-level atom and an actively stabilized, tunable ring dye laser acted as the driving field. Typically, it was possible to achieve atomic number densities of 10^{10} atoms-cm $^{-3}$ and a driving-field Rabi frequency of 2.6×10^9 s $^{-1}$. Of paramount concern was the possibility that phase-matched multi-wave mixing processes could contribute a significant number of photons to the resonator and thus obscure two-photon lasing. In addition, broadening mechanisms that might reduce the two-photon gain, such as Doppler and collisional broadening, needed to be removed. Suppression of these effects was accomplished using a collimated, small-diameter (~ 2 mm) atomic beam of atoms that passed through the center of a high-finesse resonator where the atomic-beam, the cavity and the driving-field axes were mutually orthogonal, as shown in Fig. 11. The laser resonator and the atomic beam source were housed in a high-vacuum chamber to maintain the low-loss characteristics of the resonator (0.15% per pass). Note that the threshold parameters for the two-photon laser studied in the experiment are somewhat different from the ones used in the previous discussion because: the atomic constants are slightly different; there is only natural broadening; the resonator has lower loss; and the atoms only fill a small fraction of the resonator. Accounting for these effects, we find that $B^{(2)} \sim 3 \times 10^{-2}$ s $^{-1}$, $q_{sat} = 4.5 \times 10^4$ photons (corresponding to $I_{sat} = 0.8$ W-cm $^{-2}$), and $\Delta N_o^{th} = 1 \times 10^9$ atoms-cm $^{-3}$ for $\Omega'_d = 2.7 \times 10^9$ s $^{-1}$.

Several preliminary experiments were performed to characterize the dressed-atom laser system. In the first experiment, the optical power emitted out one end of the cavity was measured as the cavity resonance frequency ω_c was tuned through the dressed-atom resonances. It was found that normal one-photon dressed-atom lasing occurred when $\omega_c \approx \omega_d - \Omega'_d$ and that no lasing occurred when $\omega_c \approx \omega_d - \Omega'_d/2$ (the predicted frequency of maximum two-photon gain). This measurement confirmed that the one-photon gain feature was spectrally removed from the region where high two-photon gain was expected and hence it would not obscure two-photon lasing. The next experiment measured directly the gain of the system by observing the transmission of an intense laser beam (approximately matched to the fundamental cavity mode and resonant with it) through the cavity. A pronounced, intensity-dependent gain feature was observed when $\omega_c \approx \omega_d - \Omega'_d/2$ which coincides precisely with the predicted location of the two-photon gain feature.

Based on the discussion of the previous sections, two-photon lasing should not occur in this system until both threshold conditions are satisfied. To determine whether there was a sufficient inversion density in the resonator to support two-photon lasing, ω_c was

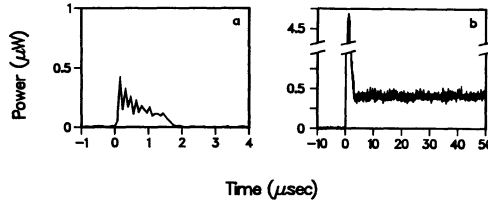


Figure 12 Time-resolved power emitted from the cavity before and after a 1.4- μsec trigger pulse is injected into the cavity. (a) The injected pulse is insufficient to satisfy the threshold condition. (b) The two-photon laser turns on.

adjusted to the peak of the two-photon gain and trigger pulses (duration $\sim 1.4 \mu\text{sec}$) resonant with the cavity were injected into it. The time-resolved cavity output power just prior to and following the injected pulse was repeatedly recorded for various trigger-pulse powers. As shown in Fig. 12a, the trigger-pulse power was insufficient to initiate two-photon lasing and the cavity output power decayed to zero after the injected pulse is turned off. For higher injected powers, the cavity output power remains high after the trigger pulse is turned off as shown in Fig. 12b. The trigger-induced transition to a state of nonzero cavity output power is entirely consistent with the expected threshold behavior of a two-photon laser as discussed in Sec. II (see Fig. 4). This demonstrates conclusively that two-photon lasing was achieved.

VI. DISCUSSION AND FUTURE DIRECTIONS

We are currently setting up the apparatus to study two-photon lasers at Duke where our main research directions will be to thoroughly characterize two-photon lasers and to develop new types of two-photon gain media. There are still many open questions about the behavior of two-photon lasers that we will be addressing, such as the laser

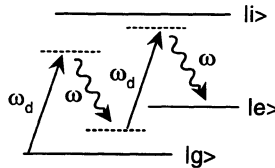


Figure 13 A new mechanism for two-photon gain: two-photon stimulated Raman scattering.

linewidth, coherence properties, instabilities in the output power, photon fluctuation noise, photon correlations and detailed studies of the threshold behavior, for example. We hope that these studies will give us a deeper understanding of the highly nonlinear interaction between light and matter and a better understanding of lasers in general.

We are searching for new two-photon gain media for two different reasons. The first reason is to have the ability to distinguish between the properties of two-photon lasers in general and those that are specific to a particular gain medium. The second reason is to make it easier to achieve two-photon lasing. The apparatus would be simplified significantly if a different source of atoms could be used, such as a vapor cell or atoms doped in a solid-state material. Toward this end, we have discovered a new two-photon gain medium that is based on a process that we call two-photon stimulated Raman scattering. The gain arises from scattering of laser photons from a three-level atom as shown in Fig. 13. We have found that the two-photon gain from this system is less susceptible to inhomogeneous broadening due to spatial variation in the pump beam intensity in comparison to the dressed-atom gain medium and that it should be possible to achieve two-photon lasing using a vapor cell. We also believe that it should be possible to develop a two-photon gain medium that is based on an analogous two-photon Brillouin process. This may pave the way for the realization of an optical fiber based two-photon laser.

Acknowledgments

We gratefully acknowledge collaborations on this work with Prof. T.W. Mossberg, Prof. M. Lewenstein, Prof. Y. Zhu, Dr. Q. Wu, S. Morin and fruitful discussions with R.W. Boyd and M.G. Raymer. This work was supported by the U.S. Army Research Office under Contract No. DAAL03-92-G-0286 and the National Science Foundation under Grant No. PHY-9357234.

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